Precalculus — Algebra I

Version 4 $- \epsilon$

by

Carl Stitz, Ph.D.
Lakeland Community College

Jeff Zeager, Ph.D.
Lorain County Community College

$\sigma o \phi \acute{\imath} \alpha$

© 2025 Carl Stitz and Jeff Zeager
https://stitz-zeager.com/

This work is licensed under CC BY-NC-SA 3.0. To view a copy of this licence, visit:

https://creativecommons.org/licenses/by-nc-sa/3.0/deed.en

Sophia Publishing
CHENNAI
https://sophia-publishing.github.io/

Table of Contents

Preface v

1 Introduction to Functions 1
- 1.1 Functions and their Representations 1
- 1.2 Constant and Linear Functions 2
- 1.3 Absolute Value Functions 3
- 1.4 Quadratic Functions 4

2 Polynomial Functions 5
- 2.1 Graphs of Polynomial Functions 5
- 2.2 The Remainder and Factor Theorems 6
- 2.3 Real Zeros of Polynomials 7
- 2.4 Complex Zeros and the Fundamental Theorem of Algebra 8

3 Rational Functions 9
- 3.1 Introduction to Rational Functions 9
- 3.2 Graphs of Rational Functions 10
- 3.3 Inequalities involving Rational Functions and Applications 11

4 Root, Radical and Power Functions 13
- 4.1 Root and Radical Functions 13
- 4.2 Power Functions 14
- 4.3 Equations and Inequalities involving Power Functions . . 15

5 Further Topics on Functions **17**
5.1 Graphs of Functions 17
5.2 Function Arithmetic 18
5.3 Function Composition 19
5.4 Transformations of Graphs 20
5.5 Relations and Implicit Functions 21
5.5.1 Exercises 40
5.5.2 Answers 45
5.6 Inverse Functions 59

Preface

Thank you for your interest in our book, but more importantly, thank you for taking the time to read the Preface. I always read the Prefaces of the textbooks which I use in my classes because I believe it is in the Preface where I begin to understand the authors - who they are, what their motivation for writing the book was, and what they hope the reader will get out of reading the text. Pedagogical issues such as content organization and how professors and students should best use a book can usually be gleaned out of its Table of Contents, but the reasons behind the choices authors make should be shared in the Preface. Also, I feel that the Preface of a textbook should demonstrate the authors' love of their discipline and passion for teaching, so that I come away believing that they really want to help students and not just make money. Thus, I thank my fellow Preface-readers again for giving me the opportunity to share with you the need and vision which guided the creation of this book and passion which both Carl and I hold for Mathematics and the teaching of it.

Carl and I are natives of Northeast Ohio. We met in graduate school at Kent State University in 1997. I finished my Ph.D in Pure Mathematics in August 1998 and started teaching at Lorain County Community College in Elyria, Ohio just two days after graduation. Carl earned his Ph.D in Pure Mathematics in August 2000 and started teaching at Lakeland Community College in Kirtland, Ohio that same month. Our schools are fairly similar in size and mission and each serves a similar population of students. The students range in age from about 16 (Ohio has a Post-Secondary Enrollment Option program which allows high school students to take college

courses for free while still in high school.) to over 65. Many of the "non-traditional" students are returning to school in order to change careers. A majority of the students at both schools receive some sort of financial aid, be it scholarships from the schools' foundations, state-funded grants or federal financial aid like student loans, and many of them have lives busied by family and job demands. Some will be taking their Associate degrees and entering (or re-entering) the workforce while others will be continuing on to a four-year college or university. Despite their many differences, our students share one common attribute: they do not want to spend $200 on a College Algebra book.

The challenge of reducing the cost of textbooks is one that many states, including Ohio, are taking quite seriously. Indeed, state-level leaders have started to work with faculty from several of the colleges and universities in Ohio and with the major publishers as well. That process will take considerable time so Carl and I came up with a plan of our own. We decided that the best way to help our students right now was to write our own College Algebra book and give it away electronically for free. We were granted sabbaticals from our respective institutions for the Spring semester of 2009 and actually began writing the textbook on December 16, 2008. Using an open-source text editor called TexNicCenter and an open-source distribution of LaTeX called MikTex 2.7, Carl and I wrote and edited all of the text, exercises and answers and created all of the graphs (using Metapost within LaTeX) for Version $0.\overline{9}$ in about eight months. (We choose to create a text in only black and white to keep printing costs to a minimum for those students who prefer a printed edition. This somewhat Spartan page layout stands in sharp relief to the explosion of colors found in most other College Algebra texts, but neither Carl nor I believe the four-color print adds anything of value.) I used the book in three sections of College Algebra at Lorain County Community College in the Fall of 2009 and Carl's colleague, Dr. Bill Previts, taught a section of College Algebra at Lakeland with the book that semester as well. Students had the option of downloading the book as a .pdf file from our website www.stitz-zeager.com[1] or buying a low-cost printed version from our colleges' respective bookstores. (By

[1] http://www.stitz-zeager.com

giving this book away for free electronically, we end the cycle of new editions appearing every 18 months to curtail the used book market.) During Thanksgiving break in November 2009, many additional exercises written by Dr. Previts were added and the typographical errors found by our students and others were corrected. On December 10, 2009, Version $\sqrt{2}$ was released. The book remains free for download at our website and by using Lulu.com[2] as an on-demand printing service, our bookstores are now able to provide a printed edition for just under $19. Neither Carl nor I have, or will ever, receive any royalties from the printed editions. As a contribution back to the open-source community, all of the LaTeX files used to compile the book are available for free under a Creative Commons License on our website as well. That way, anyone who would like to rearrange or edit the content for their classes can do so as long as it remains free.

The only disadvantage to not working for a publisher is that we don't have a paid editorial staff. What we have instead, beyond ourselves, is friends, colleagues and unknown people in the open-source community who alert us to errors they find as they read the textbook. What we gain in not having to report to a publisher so dramatically outweighs the lack of the paid staff that we have turned down every offer to publish our book. (As of the writing of this Preface, we've had three offers.) By maintaining this book by ourselves, Carl and I retain all creative control and keep the book our own. We control the organization, depth and rigor of the content which means we can resist the pressure to diminish the rigor and homogenize the content so as to appeal to a mass market. A casual glance through the Table of Contents of most of the major publishers' College Algebra books reveals nearly isomorphic content in both order and depth. Our Table of Contents shows a different approach, one that might be labeled "Functions First." To truly use The Rule of Four, that is, in order to discuss each new concept algebraically, graphically, numerically and verbally, it seems completely obvious to us that one would need to introduce functions first. (Take a moment and compare our ordering to the classic "equations first, then the Cartesian Plane and THEN functions" approach seen in most of

[2] http://www.lulu.com/content/paperback-book/college-algebra/7513097

the major players.) We then introduce a class of functions and discuss the equations, inequalities (with a heavy emphasis on sign diagrams) and applications which involve functions in that class. The material is presented at a level that definitely prepares a student for Calculus while giving them relevant Mathematics which can be used in other classes as well. Graphing calculators are used sparingly and only as a tool to enhance the Mathematics, not to replace it. The answers to nearly all of the computational homework exercises are given in the text and we have gone to great lengths to write some very thought provoking discussion questions whose answers are not given. One will notice that our exercise sets are much shorter than the traditional sets of nearly 100 "drill and kill" questions which build skill devoid of understanding. Our experience has been that students can do about 15-20 homework exercises a night so we very carefully chose smaller sets of questions which cover all of the necessary skills and get the students thinking more deeply about the Mathematics involved.

Critics of the Open Educational Resource movement might quip that "open-source is where bad content goes to die," to which I say this: take a serious look at what we offer our students. Look through a few sections to see if what we've written is bad content in your opinion. I see this open-source book not as something which is "free and worth every penny", but rather, as a high quality alternative to the business as usual of the textbook industry and I hope that you agree. If you have any comments, questions or concerns please feel free to contact me at jeff@stitz-zeager.com or Carl at carl@stitz-zeager.com.

Jeff Zeager
Lorain County Community College
January 25, 2010

Chapter 1

Introduction to Functions

1.1 Functions and their Representations

1.2 Constant and Linear Functions

1.3 Absolute Value Functions

1.4 Quadratic Functions

Chapter 2

Polynomial Functions

2.1 Graphs of Polynomial Functions

2.2 The Remainder and Factor Theorems

2.3 Real Zeros of Polynomials

2.4 Complex Zeros and the Fundamental Theorem of Algebra

Chapter 3

Rational Functions

3.1 Introduction to Rational Functions

3.2 Graphs of Rational Functions

3.3 Inequalities involving Rational Functions and Applications

Chapter 4

Root, Radical and Power Functions

4.1 Root and Radical Functions

4.2 Power Functions

4.3 Equations and Inequalities involving Power Functions

Chapter 5

Further Topics on Functions

5.1 Graphs of Functions

5.2 Function Arithmetic

5.3 Function Composition

5.4 Transformations of Graphs

5.5 Relations and Implicit Functions

Up until now in this text, we have been exclusively special kinds of mappings called *functions*. In this section, we broaden our horizons to study more general mappings called *relations*. The reader is encouraged to revisit Definition **??** in Section **??** before proceeding with the definition of *relation* below.

Definition 5.5.1. Given two sets A and B, a **relation** from A to B is a process by which elements of A are matched with (or 'mapped to') elements of B.

Unlike Definition **??**, Definition 5.5.1 puts no conditions on the process which maps elements of A to elements of B. This means that while all functions are relations, not all relations need be functions. For example, consider the mappings f and g in Figure 5.5.1 and Figure 5.5.2 from Section **??**.

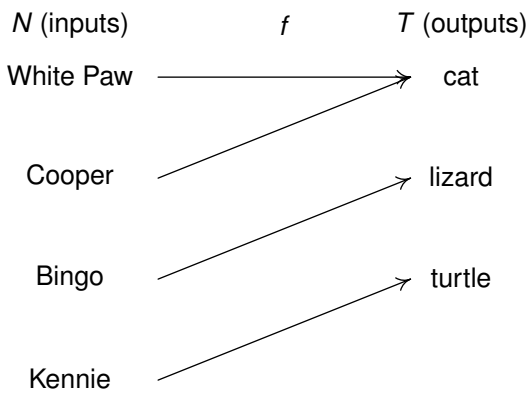


Figure 5.5.1

Both f and g are relations. More specifically, f is a *function* from N to T while g is merely *relation* from T to N. As with functions, we may describe general relations in a variety of different ways: verbally, as mapping diagrams, or a set of ordered pairs. For example, just as we may describe

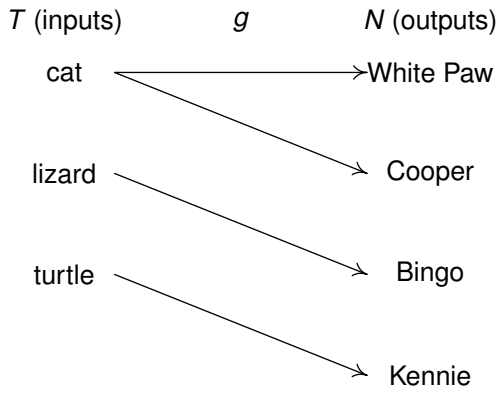


Figure 5.5.2

the function f above as

$$f = \{(\text{White Paw}, \text{cat}), (\text{Cooper}, \text{cat}), (\text{Bingo}, \text{lizard}), (\text{Kennie}, \text{turtle})\},$$

we may represent g as

$$g = \{(\text{cat}, \text{White Paw}), (\text{cat}, \text{Cooper}), (\text{lizard}, \text{Bingo}), (\text{turtle}, \text{Kennie})\}.$$

Note here the grammar 'g is a relation *from T to N*' is evidenced by the elements of T being listed first in the ordered pairs (i.e., the abscissae) and the elements of N being listed second (i.e., the ordinates.)

Unlike functions, we do not use function notation when describing the input/output relationship for general relations. For example, we may write 'f(White Paw) = cat' since f maps the input 'White Paw' to only one output, 'cat.' However, g(cat) is ambiguous since it could mean 'White Paw' or 'Cooper.'[1]

[1] In more advanced texts, we would write 'cat g White Paw' and 'cat g Cooper' to indicate g maps 'cat' to both 'White Paw' and 'Cooper.' Our study of relations, however, isn't deep enough to necessitate introducing and using this notation. Similarly, we won't introduce the notions of 'domain,' 'codomain,' and 'range' for relations, either.

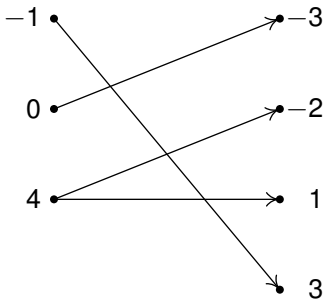


Figure 5.5.3: A Mapping Diagram of R

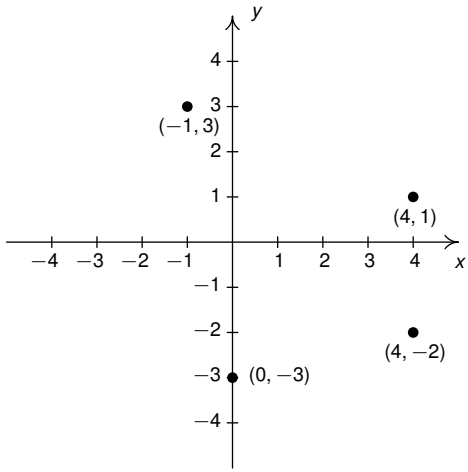


Figure 5.5.4: The graph of R

As with functions, our focus in this course will rest with relations of real numbers. Consider the relation R described as follows:

$$R = \{(-1, 3), (0, -3), (4, -2), (4, 1)\}$$

Figure 5.5.3 shows a mapping diagram of R. However, since R relates real numbers, we can also create the graph of R in the same way we graphed functions—by interpreting the ordered pairs which comprise R as points in the plane. Since we have no context, we use the default labels 'x' for the horizontal axis and 'y' for the vertical axis. See Figure 5.5.8.

Our next example focuses on using relations to describe sets of points in the plane and vice-versa.

Example 5.5.1.

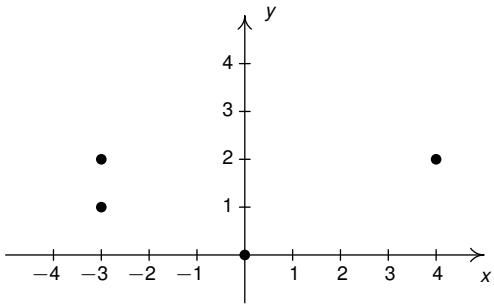
1. Graph the following relations.

   (a) $S = \{(k, 2^k) \mid k = 0, \pm 1, \pm 2\}$
   (b) $P = \{(j, j^2) \mid j \text{ is an integer}\}$

(c) $V = \{(3, y) \mid y \text{ is a real number}\}$
(d) $R = \{(x, y) \mid x \text{ is a real number}, 1 < y \leq 3\}$

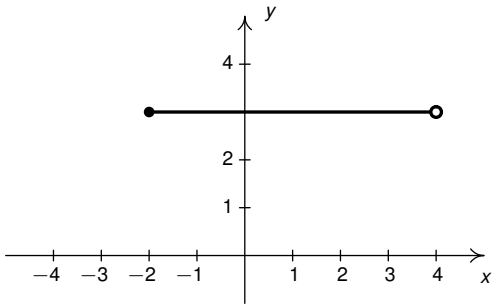
2. Find a roster or set-builder description for each of the relations below.

(a)



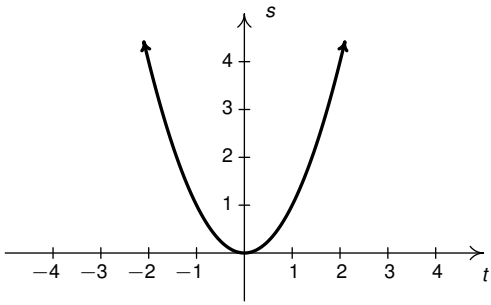
The graph of A

(b)



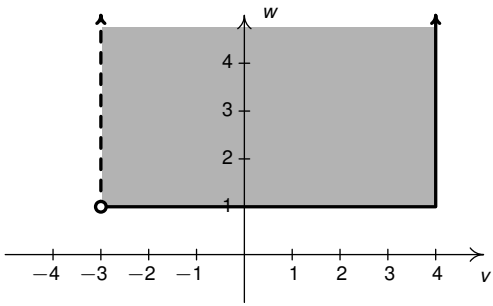
The graph of H

(c)



The graph of Q

(d)



The graph of T

Solution.

1. (a) The relation S is described using *set-builder notation*.[2] To generate the ordered pairs which belong to S, we substitute the given values of k, $k = 0, \pm 1, \pm 2$, into the formula $\left(k, 2^k\right)$.

   Starting with $k = 0$, we get $\left(0, 2^0\right) = (0, 1)$. For $k = 1$, we get $\left(1, 2^1\right) = (1, 2)$, and for $k = -1$, we get $\left(-1, 2^{-1}\right) = \left(-1, \frac{1}{2}\right)$. Continuing, we get $\left(2, 2^2\right) = (2, 4)$ for $k = 2$ and, finally $\left(-2, 2^{-2}\right) = \left(-2, \frac{1}{4}\right)$ for $k = -2$. Hence, a roster description of S is $S = \left\{\left(-2, \frac{1}{4}\right), \left(-1, \frac{1}{2}\right), (0, 1), (1, 2), (2, 4)\right\}$.

   When we graph S, we label the horizontal axis as the k-axis, since 'k' was the variable chosen used to generate the ordered

[2] See Section **??** to review this, if needed.

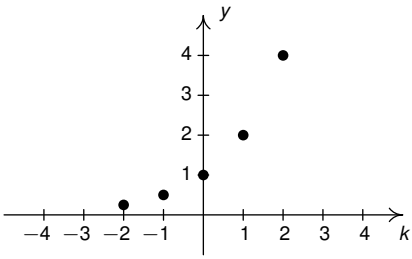


Figure 5.5.5: The graph of S

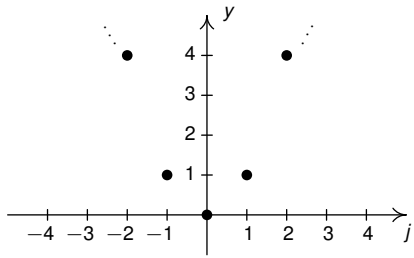


Figure 5.5.6: The graph of P

pairs and keep the default label 'y' for the vertical axis. The graph of S is given in Figure 5.5.5.

(b) To graph the relation $P = \{(j, j^2) \mid j \text{ is an integer}\}$, we proceed as above when we graphed the relation S. Here, j is restricted to being an integer, which means $j = 0, \pm 1, \pm 2$, etc.

Plugging in these sample values for j, we obtain the ordered pairs $(0, 0)$, $(1, 1)$, $(-1, 1)$, $(2, 4)$, $(-2, 4)$, etc. Since the variable j takes on only integer values, we could write P using the roster notation: $P = \{(0, 0), (\pm 1, 1), (\pm 2, 4), \ldots\}$.

We plot a few of these points and use some periods of ellipsis to indicate the complete graph contains additional points not in the current field of view. The graph of P is given in Figure 5.5.6.

(c) Next, we come to the relation V, described, once again, using set-builder notation. In this case, V consists of all ordered pairs of the form $(3, y)$ where y is free to be whatever real number we like, without any restriction.[3] For example, $(3, 0)$, $(3, -1)$, and $(3, 117)$ all belong to V as do $\left(3, \frac{1}{2}\right)$, $(3, -1.0342)$, $(3, \sqrt{2})$, etc.

After plotting some sample points, becomes apparent that the ordered pairs which belong to V correspond to points which lie on the vertical line $x = 3$, and vice-versa. That is, every point

[3] We'll revisit the concept of a 'free variable' in Section **??**.

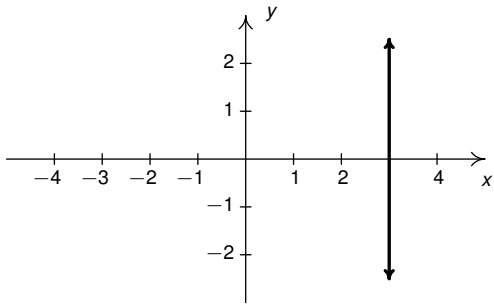


Figure 5.5.7: The graph of V

on the line $x = 3$ has coordinates which correspond to an ordered pair belonging to V. The graph of V is in Figure 5.5.7.

(d) In the relation $R = \{(x, y) \mid 1 < y \leq 3\}$, we see y is restricted by the inequality $1 < y \leq 3$, but x is free to be whatever it likes.

Since x is unrestricted, this means whatever the graph of R is, it will extend indefinitely off to the right and left. The restriction $y > 1$ means all points on the graph of R have a y-coordinate larger than one, so they are *above* the horizontal line $y = 1$. The restriction $y \leq 3$, on the other hand, means all the points on the graph of R have a y-coordinate less than or equal to 3, meaning they are either *on* or *below* the horizontal line $y = 3$.

In other words, the graph of R is the region in the plane between $y = 1$ and $y = 3$, including $y = 3$ but not $y = 1$. We signify this by *shading* the region between these two horizontal lines.

How do we communicate $y = 1$ is not part of the graph? One way is to visualize putting 'holes' all along the line $y = 1$ to indicate this is not part of the graph. In practice, however, this looks cluttered and could be confusing. Instead, we 'dash' the line $y = 1$ as seen in Figure 5.5.8.

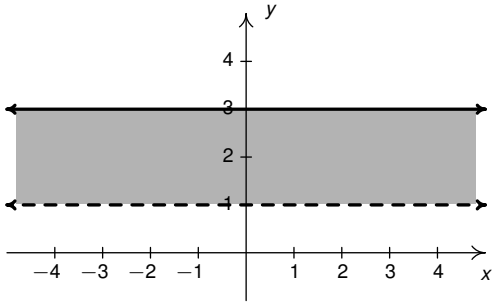


Figure 5.5.8: The graph of R

2. (a) Since A consists of finitely many points, we can describe A using the roster method:

$$A = \{(-3, 2), (-3, 1), (0, 0), (4, 2)\}.$$

(b) The graph of H appears to be a portion of the horizontal line $y = 3$ from $x = -2$ (including $x = -2$) up to, but not including $x = 4$. Since it is impossible[4] to *list* each and every one of these points, we'll opt to describe H using set-builder as opposed to the roster method. Taking a cue from the description of the relations V and R above, we write $H = \{(x, 3) \mid -2 \leq x < 4\}$.

(c) The graph of Q appears to be the graph of the function $s = f(t) = t^2$. Again, as the graph consists of infinitely many points, we will use set-builder notation to describe Q out of necessity.

There are a couple of different ways to do this. Taking a cue from the relation P above, we could write $Q = \{(t, t^2) \mid t$ is a real number Alternatively, we could introduce the dependent variable, s into the description by writing $Q = \{(t, s) \mid s = t^2\}$ where here the assumption is x takes in all real number values.

[4] Really impossible. The interested reader is encouraged to research **countable**[5] versus **uncountable**[6] sets.

(d) As with the relation R above, the relation T describes a region in the plane, The v-values appear to range between -3 (not including -3) and up to, and including, $v = 4$. The only restriction on the w-values is that $w \geq 1$, so we have $T = \{(v, w) \mid -3 < v \leq 4,\ w \geq 1\}$. □

As with functions, we can describe relations algebraically using equations. For example, the equation $v^2 + w^3 = 1$ relates two variables v and w each of which represent real numbers. More formally, we can express this sentiment by defining the relation $R = \{(v, w) \mid v^2 + w^3 = 1\}$. An ordered pair $(v, w) \in R$ means v and w are *related* by the equation $v^2 + w^3 = 1$; that is, the pair (v, w) *satisfy* the equation.

For example, to show $(3, -2) \in R$, we check that when we substitute $v = 3$ and $w = -2$, the equation $v^2 + w^3 = 1$ is true. Sure enough, $(3)^2 + (-2)^3 = 9 - 8 = 1$. Hence, R maps 3 to -2. Note, however, that $(-2, 3) \notin R$ since $(-2)^2 + (3)^3 = -8 + 27 \neq 1$ which means R does not map -2 to 3.

When asked to 'graph the equation' $v^2 + w^3 = 1$, we really have two options. We could graph the relation R above. In this case, we would be graphing $v^2 + w^3 = 1$ on the vw-plane.[7] Alternatively, we could define $S = \{(w, v) \mid v^2 + w^3 = 1\}$ and graph S. This is equivalent to graphing $v^2 + w^3 = 1$ on the wv-plane. We do both in our next example.

Example 5.5.2. Graph the equation $v^2 + w^3 = 1$ in the vw- and wv-planes. Include the axis-intercepts.

Solution.

- *graphing in the vw-plane:* We begin by finding the axis intercepts of the graph. To obtain a point on the v-axis, we require $w = 0$. To see if we have any v-intercepts on the graph of the equation $v^2 + w^3 = 1$, we substitute $w = 0$ into the equation and solve for v: $v^2 + (0)^3 = 1$. We get $v^2 = 1$ or $v = \pm 1$ so our two v-intercepts, as described in the vw-plane, are $(1, 0)$ and $(-1, 0)$.

[7] Recall this means the horizontal axis is labeled 'v' and the vertical axis is labeled 'w.'

v	w	(v, w)
-3	-2	$(-3, -2)$
-2	$-\sqrt[3]{3}$	$(-2, -\sqrt[3]{3})$
-1	0	$(-1, 0)$
0	1	$(0, 1)$
1	0	$(1, 0)$
2	$-\sqrt[3]{3}$	$(2, -\sqrt[3]{3})$
3	-2	$(3, -2)$

Table 5.5.1

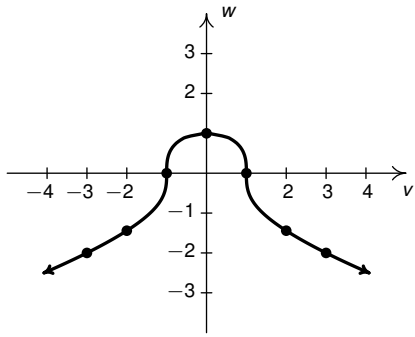


Figure 5.5.9: $v^2 + w^3 = 1$

Likewise, to find w-intercepts of the graph, we substitute $v = 0$ into the equation $v^2 + w^3 = 1$ and get $w^3 = 1$ or $w = 1$. Hence, he have only one w-intercept, $(0, 1)$.

One way to efficiently produce additional points is to solve the equation $v^2 + w^3 = 1$ for one of the variables, say w, in terms of the other, v. In this way, we are treating w as the dependent variable and v as the independent variable. From $v^2 + w^3 = 1$, we get $w^3 = 1 - v^2$ or $w = \sqrt[3]{1 - v^2}$.

We now substitute a value in for v, determine the corresponding value w, and plot the resulting point (v, w). We summarize our results in Table 5.5.1. By plotting additional points (or getting help from a graphing utility), we produce the graph in Figure 5.5.9.

- *graphing in the wv-plane:* To graph $v^2 + w^3 = 1$ in the wv-plane, all we need to do is reverse the coordinates of the ordered pairs we obtained for our graph in the vw-plane. In particular, the v-intercepts are written $(0, 1)$ and $(0, -1)$ and the w-intercept is written $(1, 0)$. Using Table 5.5.2 we produce the graph in Figure 5.5.10. □

Note that regardless of which geometric depiction we choose for $v^2 + w^3 = 1$, the graph appears to be symmetric about the w-axis. To prove this is the case, consider a generic point (v, w) on the graph of $v^2 + w^3 = 1$ in the

v	w	(w, v)
-3	-2	$(-2, -3)$
-2	$-\sqrt[3]{3}$	$(-\sqrt[3]{3}, -2)$
-1	0	$(0, -1)$
0	1	$(1, 0)$
1	0	$(0, 1)$
2	$-\sqrt[3]{3}$	$(-\sqrt[3]{3}, 2)$
3	-2	$(-2, 3)$

Table 5.5.2

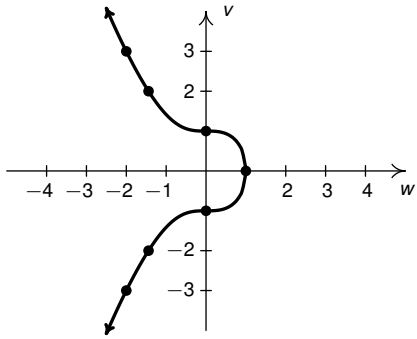


Figure 5.5.10: $v^2 + w^3 = 1$

vw-plane.

To show the point symmetric about the w-axis, $(-v, w)$ is also on the graph of $v^2 + w^3 = 1$, we need to show that the coordinates of the point $(-v, w)$ satisfy the equation $v^2 + w^3 = 1$. That is, we need to show $(-v)^2 + w^3 = 1$. Since $(-v)^2 + w^3 = v^2 + w^3$, and we know by assumption $v^2 + w^3 = 1$, we get $(-v)^2 + w^3 = v^2 + w^3 = 1$, proving $(-v, w)$ is also on the graph of the equation.

The key reason our proof above is successful is that algebraically, the equation $v^2 + w^3 = 1$ is unchanged if v is replaced with $-v$. Geometrically, this means the graph is the same if it undergoes a reflection across the w-axis. We generalize this reasoning in the following result. Note that, as usual, we default to the more common x and y-axis labels.

Theorem 5.5.1. Testing the Graph of an Equation for Symmetry:
To test the graph of an equation in the xy-plane for symmetry:

- about the x-axis: substitute $(x, -y)$ into the equation and simplify. If the result is equivalent to the original equation, the graph is symmetric about the x-axis.
- about the y-axis: substitute $(-x, y)$ into the equation and simplify. If the result is equivalent to the original equation, the graph is symmetric about the y-axis.
- about the origin: substitute $(-x, -y)$ into the equation and simplify. If the result is equivalent to the original equation, the graph is symmetric about the origin.

Parts of Theorem 5.5.1 should look familiar from our work with even and odd functions. Indeed if a function f is even, $f(-x) = f(x)$. Hence, the equation $y = f(-x)$ reduces to the equation $y = f(x)$, so the graph of f is symmetric about the y-axis.

Likewise if f is odd, then $f(-x) = -f(x)$. In this case, the equation $-y = f(-x)$ reduces to $-y = -f(x)$, or $y = f(x)$, proving the graph is symmetric about the origin.

When it comes to symmetry about the x-axis, most of the time this indicates a violation of the Vertical Line Test, which is why we haven't discussed that particular kind of symmetry until now.

We put Theorem 5.5.1 to good use in the following example.

Example 5.5.3. Graph each of the equations below in the xy-plane. Find the axis intercepts, if any, and prove any symmetry suggested by the graphs.

1. $x^2 - y^2 = 4$
2. $(x - 1)^2 + 4y^2 = 16$

Solution.

x	y	(x, y)
-4	$\pm 2\sqrt{3}$	$(-4, \pm 2\sqrt{3})$
-3	$\pm\sqrt{5}$	$(-3, \pm\sqrt{5})$
-2	0	$(-2, 0)$
2	0	$(2, 0)$
3	$\pm\sqrt{5}$	$(3, \pm\sqrt{5})$
4	$\pm 2\sqrt{3}$	$(4, \pm 2\sqrt{3})$

Table 5.5.3

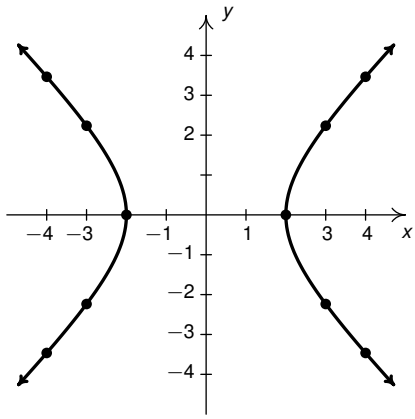


Figure 5.5.11: $x^2 - y^2 = 4$

1. We begin graphing $x^2 - y^2 = 4$ by checking for axis intercepts. To check for x-intercepts, we set $y = 0$ and solve $x^2 - (0)^2 = 4$. We get $x = \pm 2$ and obtain two x-intercepts $(-2, 0)$ and $(2, 0)$.

   When looking for y-intercepts, we set $x = 0$ and get $(0)^2 - y^2 = 4$ or $y^2 = -4$. Since this equation has no real number solutions, we have no y-intercepts.

   In order to produce more points on the graph, we solve $x^2 - y^2 = 4$ for y and obtain $y = \pm\sqrt{x^2 - 4}$. Since we know $x^2 - 4 \geq 0$ in order to produce real number results for y, we restrict our attention to $x \leq -2$ and $x \geq 2$. Doing so produces Table 5.5.3. Using these, we construct the graph in Figure 5.5.13.

   The graph certainly appears to be symmetric about both axes and the origin. To prove this, we note that the equation $x^2 - (-y)^2 = 4$ quickly reduces to $x^2 - y^2 = 4$, proving the graph is symmetric about the x-axis.

   Likewise, the equations $(-x)^2 - y^2 = 4$ and $(-x)^2 - (-y)^2 = 4$ also reduce to $x^2 - y^2 = 4$, proving the graph is, indeed, symmetric about the y-axis and origin, respectively.

y	x	(x, y)
-2	1	$(1, -2)$
-1	$1 \pm 2\sqrt{3}$	$(1 \pm 2\sqrt{3}, -1)$
0	$1 \pm 4 = -3, 5$	$(-3, 0), (5, 0)$
1	$1 \pm 2\sqrt{3}$	$(1 \pm 2\sqrt{3}, 1)$
2	1	$(1, 2)$

Table 5.5.4

2. To determine if there are any x-intercepts on the graph of $(x-1)^2 + 4y^2 = 16$, we set $y = 0$ and solve $(x-1)^2 + 4(0)^2 = 16$. This reduces to $(x-1)^2 = 16$ which gives $x = -3$ and $x = 5$. Hence, we have two x-intercepts, $(-3, 0)$ and $(5, 0)$.

   Looking for y-intercepts, we set $x = 0$ and solve $(0-1)^2 + 4y^2 = 16$ or $1 + 4y^2 = 16$. This gives $y^2 = \frac{15}{4}$ so $y = \pm\frac{\sqrt{15}}{2}$. Hence, we have two y-intercepts: $\left(0, \pm\frac{\sqrt{15}}{2}\right)$.

   In this case, it is slightly easier[8] to solve for x in terms of y. From $(x-1)^2 + 4y^2 = 16$ we get $(x-1)^2 = 16 - 4y^2$ which gives $x = 1 \pm \sqrt{16 - 4y^2}$.

   Since we know $16 - 4y^2 \geq 0$ to produce real number results for x, we require $-2 \leq y \leq 2$. Selecting values in that range produces Table 5.5.4. Plotting these points, along with the y-intercepts produces the graph in Figure 5.5.12.

   The graph certainly appears to be symmetric about the x-axis. To check, we substitute $(-y)$ in for y and get $(x-1)^2 + 4(-y)^2 = 16$ which reduces to $(x-1)^2 + 4y^2 = 16$.

   Owing to the placement of the x-intercepts, $(-3, 0)$ and $(5, 0)$, the graph is most certainly not symmetric about the y-axis nor about the origin. □

[8] Read this as we're avoiding fractions.

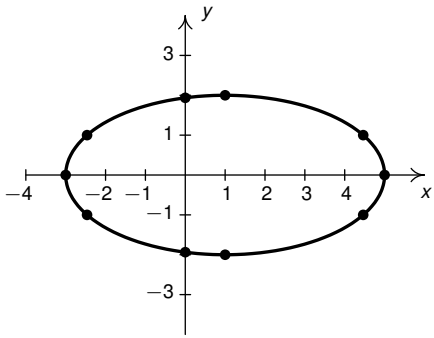


Figure 5.5.12: $(x-1)^2+4y^2=16$

Looking at the graphs of the equations $x^2-y^2=4$ and $(x-1)^2+4y^2=16$ in Example 5.5.3, it is evident neither of these equations represents y as a function of x nor x as a function of y. (Do you see why?)

With the concept of 'function' being touted in the opening remarks of Section **??** as being one of the 'universal tools' with which scientists and engineers solve a wide variety of problems, you may well wonder if we can't somehow apply what we know about functions to these sorts of relations. It turns out that while, taken all at once, these equations do not describe functions, taken in parts, they do.

For example, consider the equation $x^2-y^2=4$ (Figure 5.5.13). Solving for y, we obtained $y=\pm\sqrt{x^2-4}$. Defining $f_1(x)=\sqrt{x^2-4}$ and $f_2(x)=-\sqrt{x^2-4}$, we get a functional description for the upper and lower halves, or *branches* of the curve, respectively (autoreffig:fonexeqsqrtxsquaredminusfour and Figure 5.5.15).[9]

If, for instance, we wanted to analyze this curve near $(3,-\sqrt{5})$, we could use the *function* f_2 and all the associated function tools[10] to do just that.

[9]There are many more ways to break this relation into functional parts. We could, for instance, go piecewise and take portions of the graph which lie in Quadrants I and III as one function and leave the parts in Quadrants II and IV as the other; we could look at this as being comprised of *four* functions, and so on.

[10]including, when the time comes, Calculus

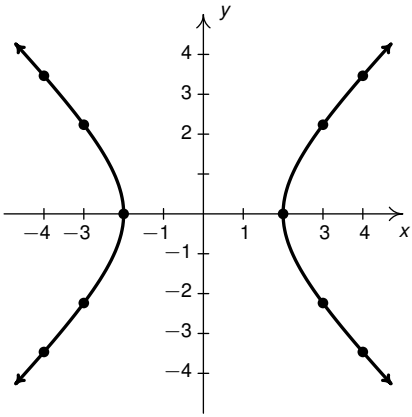


Figure 5.5.13: $x^2 - y^2 = 4$

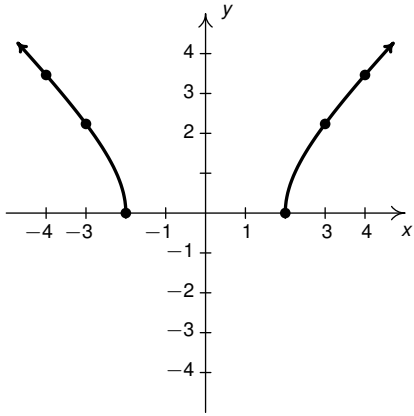


Figure 5.5.14: $f_1(x) = \sqrt{x^2 - 4}$

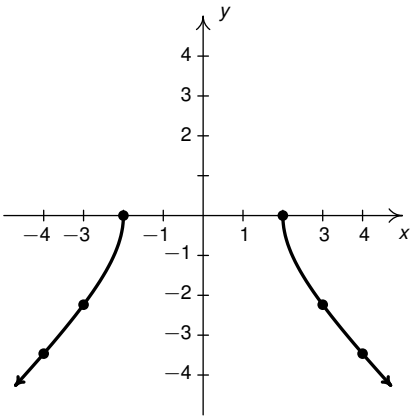


Figure 5.5.15: $f_2(x) = -\sqrt{x^2 - 4}$

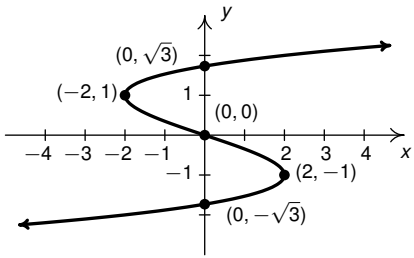


Figure 5.5.16: The graph of R

In this way we say the equation $x^2 - y^2 = 4$ *implicitly* describes y as a function of x meaning that given any point (x_0, y_0) on $x^2 - y^2 = 4$, we can find a function f defined (on an interval) containing x_0 so that $f(x_0) = y_0$ and whose graph lies on the curve $x^2 - y^2 = 4$.

Note that in this case, we are fortunate to have two *explicit* formulas for functions that cover the entire curve, namely $f_1(x) = \sqrt{x^2 - 4}$ and $f_2(x) = -\sqrt{x^2 - 4}$. We explore this concept further in the next example.

Example 5.5.4. Consider the graph of the relation R in Figure 5.5.16.

1. Explain why this curve does not represent y as a function of x.
2. Resolve the graph of R into two or more graphs of implicitly defined functions.
3. Explain why this curve represents x as a function of y and find a formula for $x = g(y)$.

Solution.

1. Using the Vertical Line Test, Theorem **??**, we find several instances where vertical lines intersect the graph of R more than once. The y-axis, $x = 0$ is one such line. We have $x = 0$ matched with *three* different y-values: $-\sqrt{3}$, 0, and $\sqrt{3}$.
2. Since the maximum number of times a vertical line intersects the graph of R is three, it stands to reason we need to resolve the graph of R into at least three pieces.

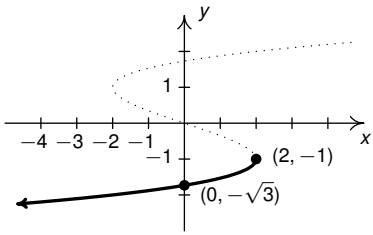


Figure 5.5.17: $y = f_1(x)$

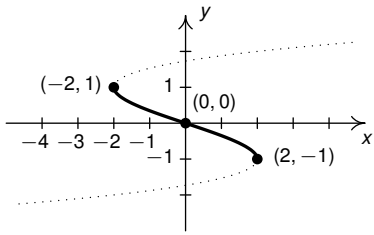


Figure 5.5.18: $y = f_2(x)$

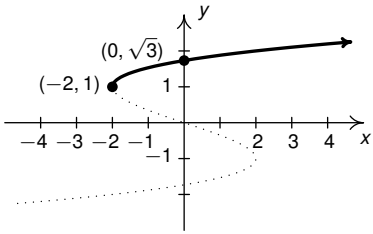


Figure 5.5.19: $y = f_3(x)$

One strategy is to begin at the far left and begin tracing the graph until it begins to 'double back' and repeat y-coordinates. Doing so we get three functions (represented by the bold solid lines) in Figure 5.5.17, Figure 5.5.18 and Figure 5.5.19.

3. To verify that R represents x as a function of y, we check to see if any y-value has more than one x associated with it. One way to do this is to employ the the Horizontal Line Test (Exercise **??** in Section **??**.) Since every horizontal line intersects the graph at most once, x is a function of y.

   Using Theorem **??** from Chapter 2, we get $x = (1)y(y - \sqrt{3})(y + \sqrt{3}) = y^3 - 3y$, a fact we can readily check using a graphing utility. □

Not all equations implicitly define y as a function of x. For a quick example, take $x = 117$ or any other vertical line. Even if an equation implicitly describes y as a function of x near one point, there's no guarantee we

can find an explicit algebraic representation for that function.[11]

While the theory of implicit functions is well beyond the scope of this text, we will nevertheless see this concept come into play in Section **??**. For our purposes, it suffices to know that just because a relation is not a function doesn't mean we cannot find a way to apply what we know about functions to analyze the relation locally through a functional lens.

[11] An example of this is $y^5 - y - x = 1$ near $(-1, 0)$.

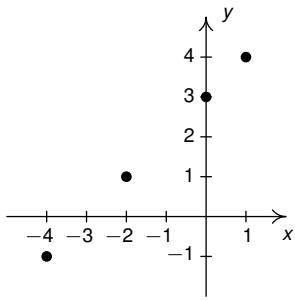
5.5.1 Exercises

In Exercises 1 - 20, graph the given relation in the xy-plane.

1. $\{(-3,9),\ (-2,4),\ (-1,1),\ (0,0),\ (1,1),\ (2,4),\ (3,9)\}$
2. $\{(-2,0),\ (-1,1),\ (-1,-1),\ (0,2),\ (0,-2),\ (1,3),\ (1,-3)\}$
3. $\{(m,2m) \mid m = 0, \pm 1, \pm 2\}$
4. $\left\{\left(\frac{6}{k}, k\right) \mid k = \pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6\right\}$
5. $\left\{\left(n, 4-n^2\right) \mid n = 0, \pm 1, \pm 2\right\}$
6. $\left\{\left(\sqrt{j}, j\right) \mid j = 0, 1, 4, 9\right\}$
7. $\{(x,-2) \mid x > -4\}$
8. $\{(x,3) \mid x \leq 4\}$
9. $\{(-1,y) \mid y > 1\}$
10. $\{(2,y) \mid y \leq 5\}$
11. $\{(-2,y) \mid -3 < y \leq 4\}$
12. $\{(3,y) \mid -4 \leq y < 3\}$
13. $\{(x,2) \mid -2 \leq x < 3\}$
14. $\{(x,-3) \mid -4 < x \leq 4\}$
15. $\{(x,y) \mid x > -2\}$
16. $\{(x,y) \mid x \leq 3\}$
17. $\{(x,y) \mid y < 4\}$
18. $\{(x,y) \mid x \leq 3,\ y < 2\}$
19. $\{(x,y) \mid x > 0,\ y < 4\}$
20. $\{(x,y) \mid -\sqrt{2} \leq x \leq \frac{2}{3},\ \pi < y \leq \frac{9}{2}\}$

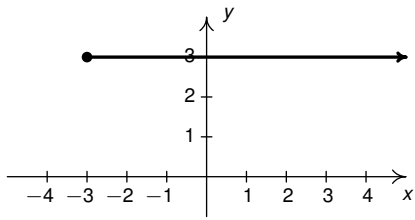
In Exercises 1. - 10., describe the given relation using either the roster or set-builder method.

1.



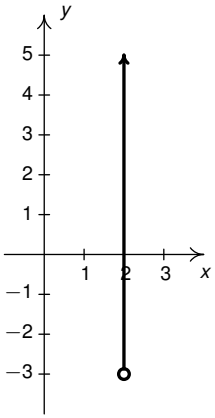
Relation A

2.



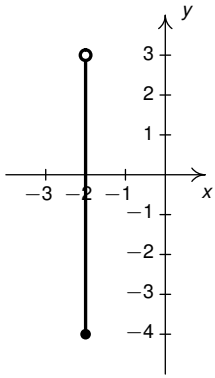
Relation B

3.



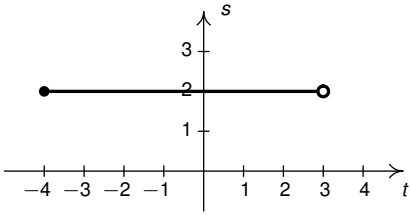
Relation *C*

4.



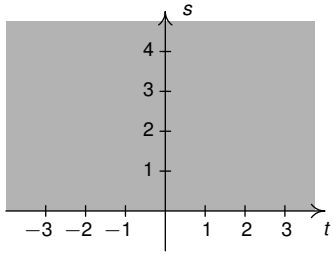
Relation *D*

5.



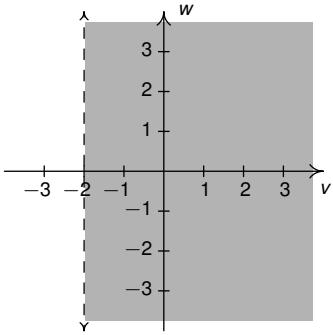
Relation *E*

6.



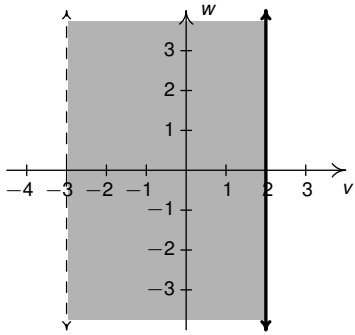
Relation *F*

7.



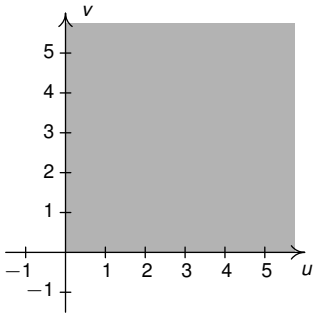
Relation *G*

8.



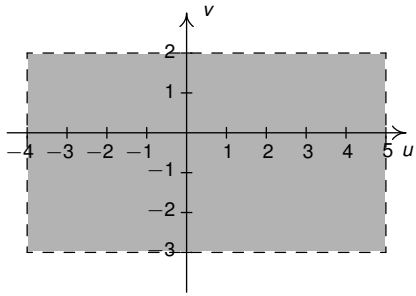
Relation *H*

9.



Relation I

10.



Relation J

Some relations are fairly easy to describe in words or with the roster method but are rather difficult, if not impossible, to graph. Discuss with your classmates how you might graph the relations given in Exercises 11 - 14. Note that in the notation below we are using the ellipsis, '. . . ,' to denote that the list does not end, but rather, continues to follow the established pattern indefinitely.

For the relations in Exercises 11 and 12, give two examples of points which belong to the relation and two points which do not belong to the relation.

11. $\{(x, y) \mid x$ is an odd integer, and y is an even integer.$\}$
12. $\{(x, 1) \mid x$ is an irrational number $\}$
13. $\{(1, 0), (2, 1), (4, 2), (8, 3), (16, 4), (32, 5), \ldots\}$
14. $\{\ldots, (-3, 9), (-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4), (3, 9), \ldots\}$

For each equation given in Exercises 15 - 18:

- Graph the equation in the xy-plane by creating a table of points.
- Find the axis intercepts, if they exist.
- Test the equation for symmetry. If the equation fails a symmetry test, find a point on the graph of the equation whose symmetric point is not on the graph of the equation.
- Determine if the equation describes y as a function of x. If not, describe the graph of the equation using two or more explicit functions of x. Check your answers using a graphing utility.

15. $(x+2)^2 + y^2 = 16$

16. $x^2 - y^2 = 1$

17. $4y^2 - 9x^2 = 36$

18. $x^3 y = -4$

For each equation given in Exercises 19 - 22:

- Graph the equation in the vw-plane by creating a table of points.
- Find the axis intercepts, if they exist.
- Test the equation for symmetry. If the equation fails a symmetry test, find a point on the graph of the equation whose symmetric point is not on the graph of the equation.
- Determine if the equation describes w as a function of v. If not, describe the graph of the equation using two or more explicit functions of v. Check your answers using a graphing utility.

19. $v + w^2 = 4$

20. $v^3 + w^3 = 8$

21. $v^2 w^3 = 8$

22.[12] $v^4 - 2v^2 w + w^2 = 16$

The procedures which we have outlined in the Examples of this section and used in Exercises 15 - 18 all rely on the fact that the equations were "well-behaved". Not everything in Mathematics is quite so tame, as the following equations will show you. Discuss with your classmates how you might approach graphing the equations given in Exercises 23 - 26. What difficulties arise when trying to apply the various tests and procedures given in this section? For more information, including pictures of the curves, each curve name is a link to its page at www.wikipedia.org. For a much longer list of fascinating curves, click here[13].

[12] HINT: $v^4 - 2v^2 w + w^2 = \left(v^2 - w\right)^2 \ldots$

[13] http://en.wikipedia.org/wiki/List_of_curves

23. $x^3 + y^3 - 3xy = 0$
Folium of Descartes[14]

24. $x^4 = x^2 + y^2$
Kampyle of Eudoxus[15]

25. $y^2 = x^3 + 3x^2$
Tschirnhausen cubic[16]

26. $(x^2 + y^2)^2 = x^3 + y^3$
Crooked egg[17]

27. With the help of your classmates, find examples of equations whose graphs possess

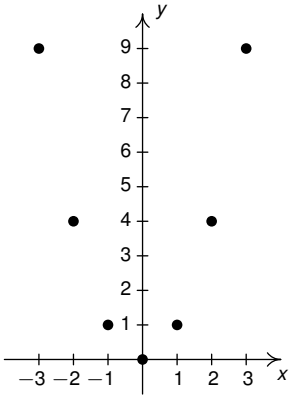
    - symmetry about the x-axis only
    - symmetry about the y-axis only
    - symmetry about the origin only
    - symmetry about the x-axis, y-axis, and origin

    Can you find an example of an equation whose graph possesses exactly *two* of the symmetries listed above? Why or why not?

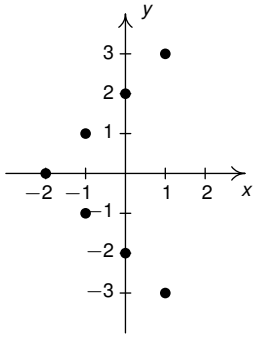
[14] http://en.wikipedia.org/wiki/Folium_of_descartes
[15] http://en.wikipedia.org/wiki/Kampyle_of_Eudoxus
[16] http://en.wikipedia.org/wiki/Tschirnhausen_cubic
[17] https://en.wikipedia.org/wiki/File:Crooked_egg_curve.svg

5.5.2 Answers

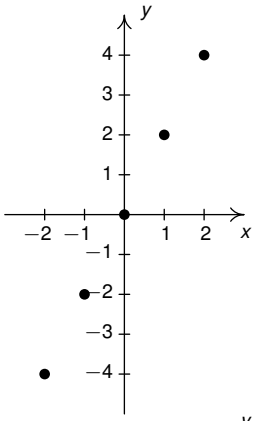
1.



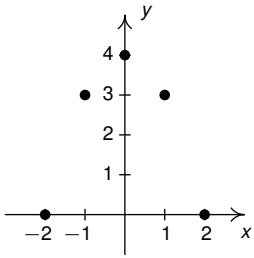
2.

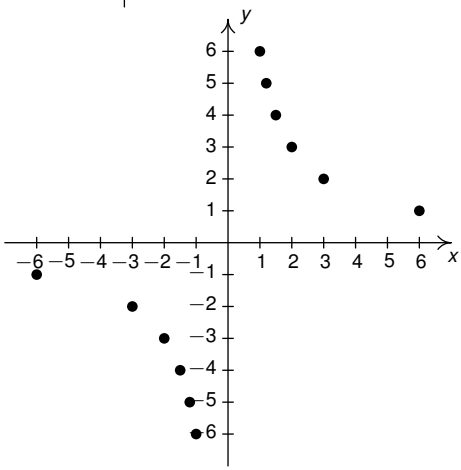


3.

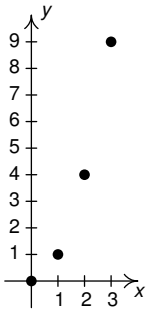

y
4
3
2
1
−2 −1 1 2 x
−1
−2
−3
−4

4.

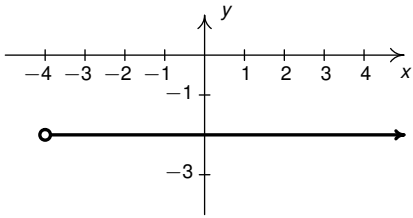

y
4
3
2
1
−2 −1 1 2 x


y
6
5
4
3
2
1
−6 −5 −4 −3 −2 −1 1 2 3 4 5 6 x
−1
−2
−3
−4
−5
−6

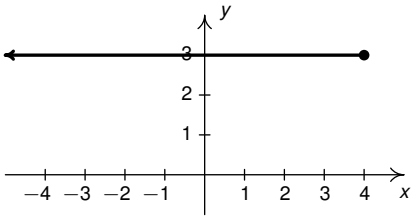
5.


y
9
8
7
6
5
4
3
2
1
1 2 3 x

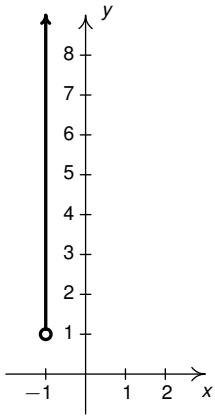
6.


y
−4 −3 −2 −1 1 2 3 4 x
−1
−3

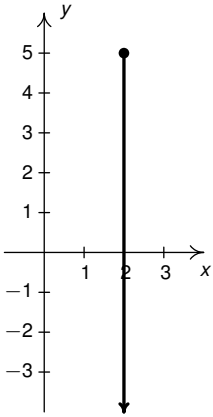
7.


y
3
2
1
−4 −3 −2 −1
1 2 3 4
x

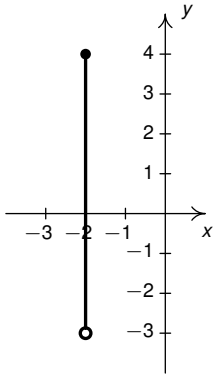
8.


y
8
7
6
5
4
3
2
1
−1
1 2
x

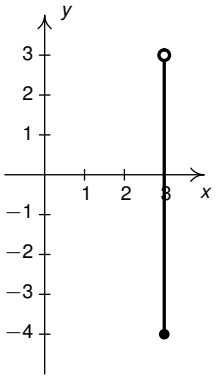
9.


y
5
4
3
2
1
1 2 3
x
−1
−2
−3

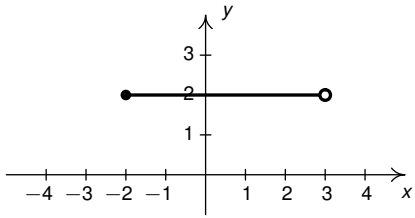
10.


y
4
3
2
1
−3 −2 −1
x
−1
−2
−3

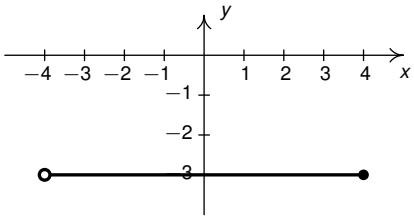
11.



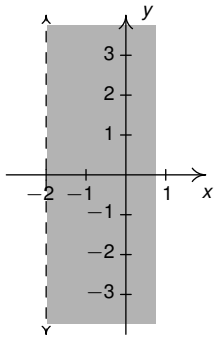
12.



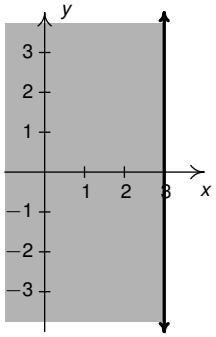
13.



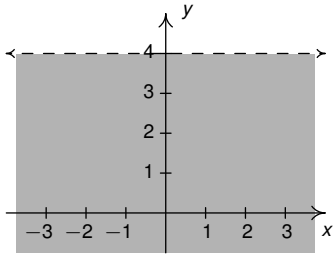
14.



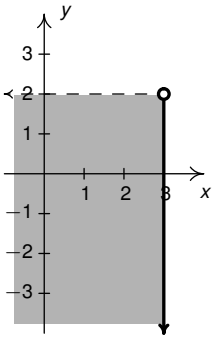
15.



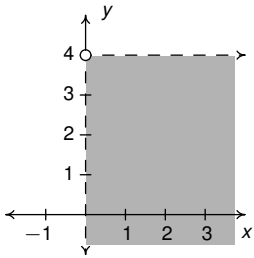
16.



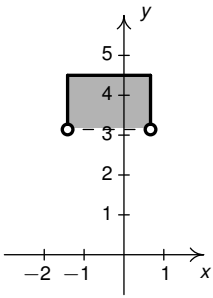
17.



18.



19.



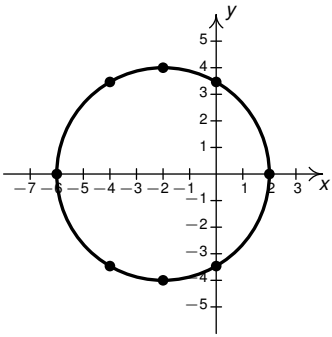
20. $A = \{(-4,-1),(-2,1),(0,3),(1,4)\}$ 21. $B = \{(x,3) \mid x \geq -3\}$
22. $C = \{(2,y) \mid y > -3\}$ 23. $D = \{(-2,y) \mid -4 \leq y < 3\}$
24. $E = \{(t,2) \mid -4 < t \leq 3\}$ 25. $F = \{(t,s) \mid s \geq 0\}$
26. $G = \{(v,w) \mid v > -2\}$ 27. $H = \{(v,w) \mid -3 < v \leq 2\}$
28. $I = \{(u,v) \mid u \geq 0, v \geq 0\}$
29. $J = \{(u,v) \mid -4 < u < 5,\ -3 < v < 2\}$

34. $(x+2)^2 + y^2 = 16$
Re-write as $y = \pm\sqrt{16-(x+2)^2}$.

x-intercepts: $(-6,0)$, $(2,0)$

y-intercepts: $\left(0, \pm 2\sqrt{3}\right)$

x	y	(x, y)
-6	0	$(-6, 0)$
-4	$\pm 2\sqrt{3}$	$\left(-4, \pm 2\sqrt{3}\right)$
-2	± 4	$(-2, \pm 4)$
0	$\pm 2\sqrt{3}$	$\left(0, \pm 2\sqrt{3}\right)$
2	0	$(2, 0)$



The graph is symmetric about the x-axis

The graph is not symmetric about the y-axis: $(-6, 0)$ is on the graph but $(6, 0)$ is not.

The graph is not symmetric about the origin: $(-6, 0)$ is on the graph but $(6, 0)$ is not.

The equation does not describe y as a function of x.

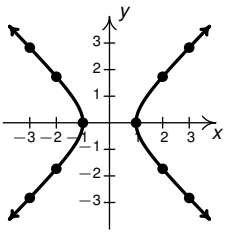
The graph of the equation is the graphs of $f_1(x) = \sqrt{16 - (x+2)^2}$ together with $f_2(x) = -\sqrt{16 - (x+2)^2}$.

35. $x^2 - y^2 = 1$
Re-write as: $y = \pm\sqrt{x^2 - 1}$.

x-intercepts: $(-1, 0), (1, 0)$

The graph has no y-intercepts

x	y	(x, y)
-3	$\pm\sqrt{8}$	$(-3, \pm\sqrt{8})$
-2	$\pm\sqrt{3}$	$(-2, \pm\sqrt{3})$
-1	0	$(-1, 0)$
1	0	$(1, 0)$
2	$\pm\sqrt{3}$	$(2, \pm\sqrt{3})$
3	$\pm\sqrt{8}$	$(3, \pm\sqrt{8})$



The graph is symmetric about the x-axis.

The graph is symmetric about the y-axis.

The graph is symmetric about the origin.

The equation does not describe y as a function of x.

The graph of the equation is the graphs of $f_1(x) = \sqrt{x^2 - 1}$ together with $f_2(x) = -\sqrt{x^2 - 1}$.

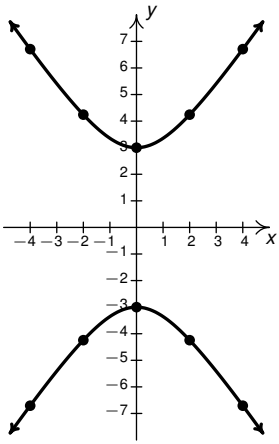
36. $4y^2 - 9x^2 = 36$

Re-write as: $y = \pm\dfrac{\sqrt{9x^2 + 36}}{2}$.

The graph has no x-intercepts

y-intercepts: $(0, \pm 3)$

x	y	(x, y)
-4	$\pm 3\sqrt{5}$	$\left(-4, \pm 3\sqrt{5}\right)$
-2	$\pm 3\sqrt{2}$	$\left(-2, \pm 3\sqrt{2}\right)$
0	± 3	$(0, \pm 3)$
2	$\pm 3\sqrt{2}$	$\left(2, \pm 3\sqrt{2}\right)$
4	$\pm 3\sqrt{5}$	$\left(4, \pm 3\sqrt{5}\right)$



The graph is symmetric about the x-axis.

The graph is symmetric about the y-axis.

The graph is symmetric about the origin.

The equation does not describe y as a function of x.

The graph of the equation is the graphs of $f_1(x) = \dfrac{\sqrt{9x^2 + 36}}{2}$ together with $f_2(x) = -\dfrac{\sqrt{9x^2 + 36}}{2}$.

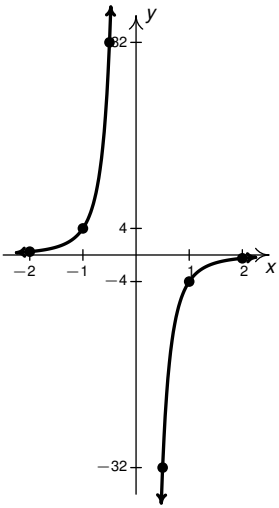
37. $x^3 y = -4$

Re-write as: $y = -\dfrac{4}{x^3} = -4x^{-3}$.

The graph has no x-intercepts

The graph has no y-intercepts

x	y	(x, y)
-2	$\frac{1}{2}$	$(-2, \frac{1}{2})$
-1	4	$(-1, 4)$
$-\frac{1}{2}$	32	$(-\frac{1}{2}, 32)$
$\frac{1}{2}$	-32	$(\frac{1}{2}, -32)$
1	-4	$(1, -4)$
2	$-\frac{1}{2}$	$(2, -\frac{1}{2})$



The graph is not symmetric about the x-axis: $(1, -4)$ is on the graph but $(1, 4)$ is not.

The graph is not symmetric about the y-axis: $(1, -4)$ is on the graph but $(-1, -4)$ is not.

The graph is symmetric about the origin.

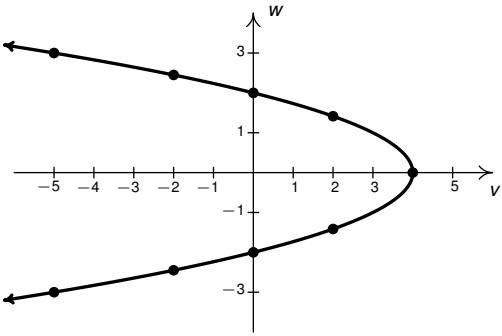
The equation does describe y as a function of x, namely $y = f(x) = -4x^{-3}$.

38. $v + w^2 = 4$
Re-write as $w = \pm\sqrt{4 - v}$.

v-intercept: $(4, 0)$

w-intercepts: $(0, \pm 2)$

v	w	(x, y)
-5	± 3	$(-5, \pm 3)$
-2	$\pm\sqrt{6}$	$\left(-2, \pm\sqrt{6}\right)$
0	± 2	$(0, \pm 2)$
2	$\pm\sqrt{2}$	$\left(1, \pm\sqrt{3}\right)$
4	0	$(4, 0)$



The graph is symmetric about the v-axis

The graph is not symmetric about the w-axis: $(4, 0)$ is on the graph but $(-4, 0)$ is not.

The graph is not symmetric about the origin: $(4, 0)$ is on the graph but $(-4, 0)$ is not.

The equation does not describe w as a function of v.

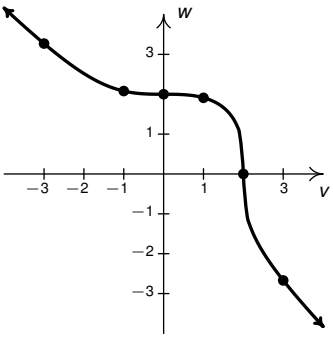
The graph of the equation is the graphs of $f_1(v) = \sqrt{4 - v}$ together with $f_2(v) = -\sqrt{4 - v}$.

39. $v^3 + w^3 = 8$
Re-write as: $w = \sqrt[3]{8 - v^3}$.

v-intercept: $(2, 0)$

w-intercept: $(0, 2)$

v	w	(v, w)
-3	$\sqrt[3]{35}$	$(-3, \sqrt[3]{35})$
-1	$\sqrt[3]{9}$	$(-1, \sqrt[3]{9})$
0	2	$(0, 2)$
1	$\sqrt[3]{7}$	$(1, \sqrt[3]{7})$
2	0	$(2, 0)$
3	$-\sqrt[3]{19}$	$(3, -\sqrt[3]{19})$



The graph is not symmetric about the v-axis: $(0, 2)$ is on the graph but $(0, -2)$ is not.

The graph is not symmetric about the w-axis: $(2, 0)$ is on the graph but $(-2, 0)$ is not.

The graph is not symmetric about the origin: $(0, 2)$ is on the graph but $(0, -2)$ is not.

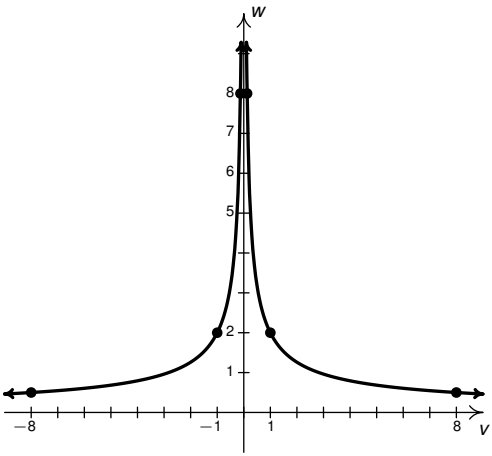
The equation does describe w as a function of v, namely $w = f(v) = \sqrt[3]{8 - v^3}$.

40. $v^2 w^3 = 8$
Re-write as $w = \dfrac{2}{\sqrt[3]{v^2}} = 2v^{-\frac{2}{3}}$.

The graph has no v-intercepts.

The graph has no w-intercepts.

v	w	(x, y)
-8	$\frac{1}{2}$	$\left(-8, \frac{1}{2}\right)$
-1	2	$(-1, 2)$
$-\frac{1}{8}$	8	$\left(-\frac{1}{8}, 8\right)$
$\frac{1}{8}$	8	$\left(\frac{1}{8}, 8\right)$
1	2	$(1, 2)$
8	$\frac{1}{2}$	$\left(8, \frac{1}{2}\right)$



The graph is not symmetric about the v-axis: $(-1, 2)$ is on the graph but $(-1, -2)$ is not.

The graph is symmetric about the w-axis.

The graph is not symmetric about the origin: $(-1, 2)$ is on the graph but $(-1, -2)$ is not.

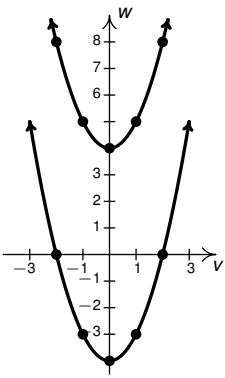
The equation does describe w as a function of v, namely $w = f(v) = 2v^{-\frac{2}{3}}$.

41. $v^4 - 2v^2w + w^2 = 16$
Re-write as: $\left(v^2 - w\right)^2 = 16$
Extracting square roots gives:
$w = v^2 + 4$ and $w = v^2 - 4$

v-intercepts: $(-2, 0), (2, 0)$.

w-intercepts: $(0, -4), (0, 4)$

v	w	(v, w)
-2	8	$(-2, 8)$
-2	0	$(-2, 0)$
-1	5	$(-1, 5)$
-1	-3	$(-1, -3)$
0	± 4	$(0, \pm 4)$
1	5	$(1, 5)$
1	-3	$(1, -3)$
2	8	$(2, 8)$
2	0	$(2, 0)$



The graph is not symmetric about the v-axis: $(1, 5)$ is on the graph but $(1, -5)$ is not.

The graph is symmetric about the w-axis.

The graph is not symmetric about the origin: $(1, 5)$ is on the graph but $(-1, -5)$ is not.

The equation does not describe w as a function of v.

The graph of the equation is the graphs of $f_1(v) = v^2 + 4$ together with $f_2(v) = v^2 - 4$.

5.6 Inverse Functions